

MANASWI KALYAN

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EDUCATION

University of Southern California

M.S. Computer Science GPA: 3.67/4.00

Aug 2025 – May 2027

Los Angeles, CA

JSS Science and Technology University

B.E. Computer Science & Business Systems GPA: 3.69/4.00

Dec 2021 – May 2025

Karnataka, India

EXPERIENCE

Student Research Worker

Apr 2026 – Present

Keck School of Medicine, USC — Laboratory of Neuro Imaging (LONI)

Los Angeles, CA

- Implemented REOrder — a Plackett–Luce policy over patch permutations trained with REINFORCE under an EMA baseline and a Gumbel-temperature curriculum — in PyTorch, extending it from 2D to 3D volumetric MRI for Alzheimer's classification on ADNI; benchmarked it against nine fixed orderings across three transformers spanning a $268\times$ capacity range (0.33M–88.4M params).
- Ran ~ 40 experiments on an $8\times$ V100 HPC cluster with subject-level splits to prevent leakage; isolated an apparent +7.5-pt gain to measurement noise via a 2×2 ablation and seed-variance analysis, tracing each architecture's null result to a distinct mechanism.
- Integrated Rotary Position Embeddings into 3D Vision Transformers for Alzheimer's classification on ADNI T1w MRI; implemented the MiNiT and NiT variants, benchmarked them against a ViT-B/16 baseline across pretraining regimes on 44K UK Biobank and 100K synthetic LDM volumes, and reached AUC 0.921 on ADNI with 0.887 and 0.813 zero-shot on the independent NIMHANS and OASIS3 cohorts.

Full Stack Web Developer Intern

Aug 2024 – Oct 2024

JupiterKing Technologies

Karnataka, India

- Developed a Spring Boot authentication service using parameterized JPA queries, with comparative login endpoints exercising it against OWASP-style injection payloads to validate secure query handling; integrated an Angular frontend in a layered controller–service–repository architecture.

Freelance Software Developer

Oct 2023 – Feb 2024

Studio Session Manager — Endurance Studios

Karnataka, India

- Sole developer of a full-stack session management platform (React, Spring Boot, PostgreSQL) serving 15+ users; translated stakeholder requirements into booking, file-sharing, and scheduling APIs that cut manual coordination by 30+ min/session, deployed via Docker at 99%+ uptime.

PROJECTS

Multimodal RAG System

github.com/manaswiii/multimodal-rag

Python · FastAPI · FAISS · PostgreSQL · LangChain · Claude API · Streamlit · Docker

- Built a production-grade retrieval system over PDFs, images (Tesseract OCR), CSVs, and source code; integrated the Claude API for grounded answers with source citations and confidence scores.
- Implemented hybrid retrieval combining dense embeddings with BM25 keyword ranking, improving relevance over vector-only search; evaluation harness tracked ROUGE and retrieval latency (~ 100 ms).

LLM-Powered Code Review Assistant

github.com/manaswiii/code-review-assistant

Python · FastAPI · PostgreSQL · SQLAlchemy · React · Claude API · GPT-4o · Docker

- Designed a provider-agnostic LLM client abstracting Anthropic and OpenAI behind one interface, swappable by environment variable with no code changes, with exponential-backoff retry on rate limits and per-request token and cost accounting.
- Built 7 REST endpoints over 3 normalized PostgreSQL tables covering snippet review, before/after diff scoring, and batch GitHub repo review with per-file error isolation; constrained-JSON prompting returns severity-tagged findings, rendered in a React frontend.

Distributed Rate Limiter & API Gateway

github.com/manaswiii/REPO-NAME

Python · FastAPI · Redis · Lua · nginx · k6 · Prometheus · Grafana · Docker

- Built a horizontally-scalable API gateway enforcing token-bucket and sliding-window rate limits over Redis-backed shared state, with limit checks run as atomic Lua scripts to eliminate cross-replica read-modify-write races.
- Under k6 load across three replicas behind nginx, in-process limiting over-admitted $\sim 3\times$ the configured rate while the Redis-backed limiter held; Prometheus and Grafana instrumentation, fail-open/fail-closed on Redis loss, pytest concurrency tests in CI.

Chain-of-Thought Faithfulness Analysis

github.com/manaswiii/cot_faithfulness

Python · Ollama · Llama 3.2 · Qwen 2.5

- Designed a 4-experiment framework (~ 15 K LLM queries) testing whether chain-of-thought causally drives predictions; CoT improved math reasoning 44–50 pp while science MCQ showed 83% early-answer consistency, indicating post-hoc rationalization.
- Identified a “Steered-But-Hidden” failure mode in which models silently follow misleading prompt hints while emitting reasoning that never acknowledges the steer, at a 16% rate on ARC-Challenge with Qwen 2.5.

TECHNICAL SKILLS

Languages

Python, Java, C/C++, SQL (PostgreSQL), JavaScript, HTML/CSS

ML & NLP

PyTorch, TensorFlow, Keras, scikit-learn, HuggingFace Transformers, sentence-transformers

ML Methods

Vision Transformers, RoPE, REINFORCE, Plackett–Luce policies, ablation & seed-variance analysis

Retrieval & LLM

FAISS, BM25, LangChain, hybrid search, Tesseract OCR, Claude API, GPT-4o, Ollama

MLOps

A/B testing, drift detection, automated retraining, model serving, Prometheus, Grafana

Backend & Infra

FastAPI, Spring Boot, Flask, REST APIs, React, PostgreSQL, Redis, Docker, nginx, Git, GitHub Actions CI, pytest, Linux